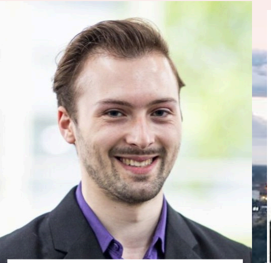
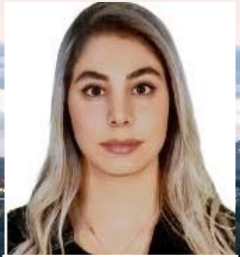


Stefan
van Laarhoven



Sotoudeh
Heidarpour

Menno Koning

Roya
Jamshidian

Koen Mulder

Kaustubh
Thakurdesai

Andi Li

Nicola Vanzetto

Introduction to CFD (4RC30)

Planning

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Dr. Yali Tang, y.tang2@tue.nl, Tel. 8052, VEC 3.106

Niels Deen

Yali Tang

Department of Mechanical Engineering

“Boundary conditions”

- **Needed knowledge: Introduction transport phenomena / Heat and flow**
- **8 lectures & 7 practice sessions with CFD code**
- **Examination:**
  - **CFD assignment (in groups of 2 or 3) → paper**
    Deadline: Sunday 21 June 2026
  - **Oral exam about paper and theory**
- **Book:**
  An Introduction to Computational Fluid Dynamics:
  The Finite Volume Method (2nd edition),
  H.K. Versteeg, W. Malalasekera

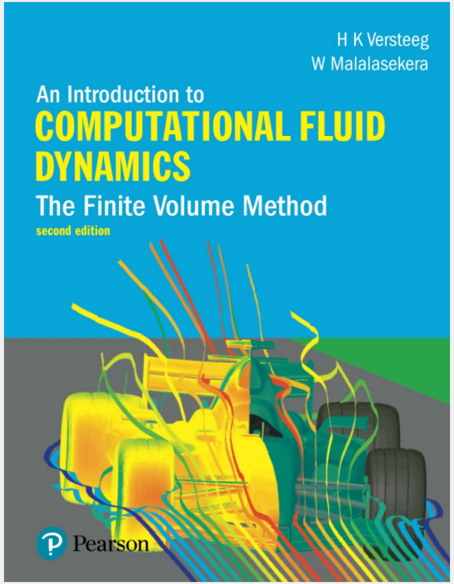


Image credit: bol.com

Timetable

- Mon 20-04-2026, 1-4 lecture 1 (ND) + tutorial 1
- Thu 23-04-2026, 5-8 lecture 2 (YT) + tutorial 2
- Mon 27-04-2026 NO lecture (King's day)
- Thu 30-04-2026, 5-8 lecture 3 (ND) + tutorial 3
- Mon 04-05-2026 NO lecture (bridge day -> TU/e closed)
- Thu 07-05-2026, 5-8 lecture 4 (YT) + tutorial 4
- Mon 11-05-2026, 1-4 lecture 5 (ND) + instruction on final assignment
- Thu 14-05-2026 NO lecture (Ascension day)
- Mon 18-05-2026, 1-4 lecture 6 (ND) + tutorial 5
- Thu 21-05-2026, 5-8 lecture 7 (YT) + tutorial 6 (start of final assignment)
- Mon 25-05-2026 NO lecture (Whit Monday)
- Thu 28-05-2026, 5-8 lecture 8 (YT) + no tutorial
- Sun 21-06-2026 DEADLINE PAPER on final assignment

ND = Niels Deen
YT = Yali Tang

Course planning

- **Lecture 1a: conservation equations** (Chapter 2)
  - Mass conservation
  - Momentum conservation
  - Internal conservation
  - General variable conservation
- **Lecture 1b: diffusion problems** (Chapter 4)
  - Discretization diffusion equation
  - Worked example: heat conduction
  - Basics of programming in Matlab
- **Lecture 2: convection-diffusion problems** (Chapter 5)
  - Discretization convection-diffusion equation
  - Discretization schemes and their properties

Course planning continued

- **Lecture 3: solving PDE's** (Chapters 6+7)
  - SIMPLE algorithm (SIMPLER, PISO: stud.)
  - TDMA
- **Lecture 4: unsteady flow problems** (Chapter 8)
  - Implicit/explicit schemes
  - Discretization
- **Lecture 5: turbulence** (Chapter 3)
  - Qualitative description of turbulence
  - Boundary layer flows
  - k-epsilon model

Course planning continued

- Lecture 6: chemical reactions (Chapter 10)
  - Simple chemical reacting system
  - Eddy break-up model
- Lecture 7: boundary conditions (Chapter 9)
  - Law of the wall, etc.
- Lecture 8: outlook
  - Advanced modelling of multiphase flows

Introduction to CFD (4RC30)

Conservation laws

Prof.dr.ir. Niels Deen, N.G.Deen@tue.nl, Tel. 3681, VEC 3.202

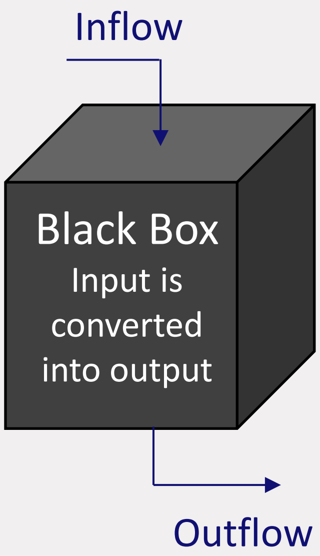
Dr. Yali Tang, y.tang2@tue.nl, Tel. 8052, VEC 3.106

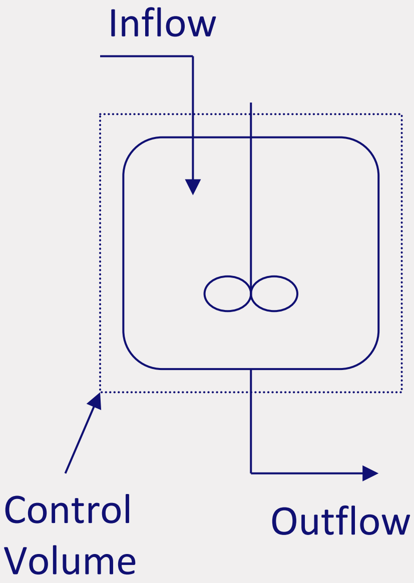
Department of Mechanical Engineering

Outline

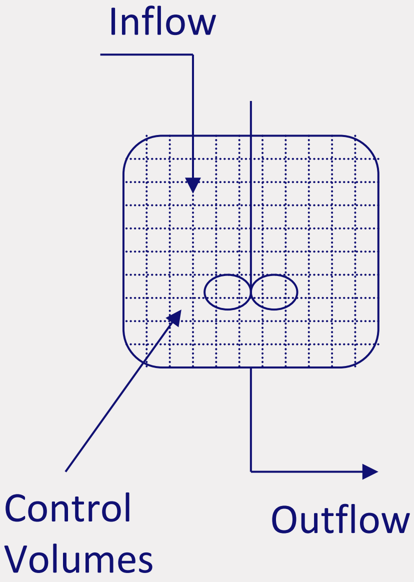
- What is CFD? Why use CFD?
- Conventions on notation
- Conservation laws
- 1D steady diffusion problems + example

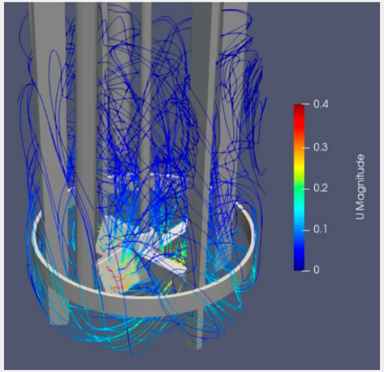
What is CFD?





zero-dimensional





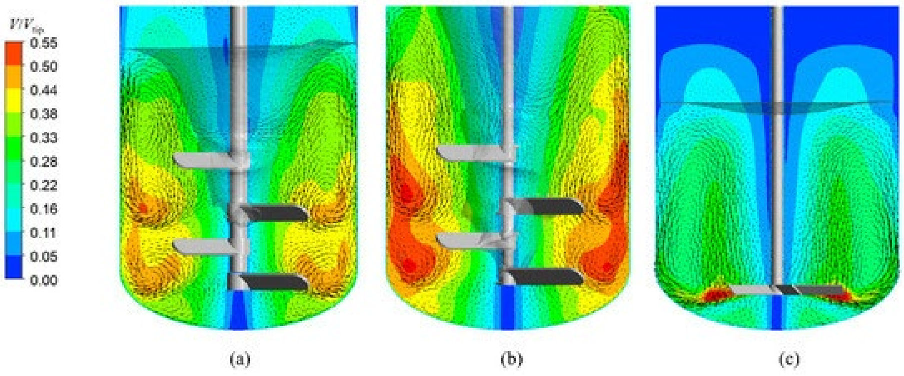
multi-dimensional

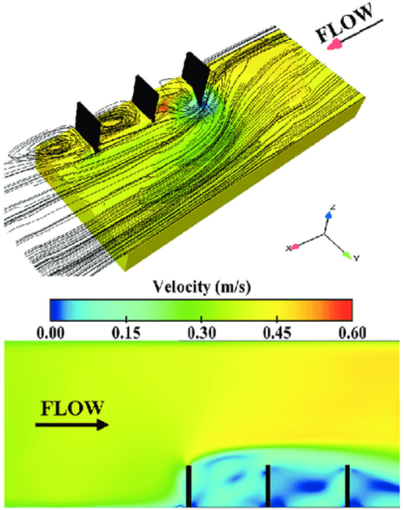
Image credit:
https://doi.org/10.3390/pr8111511

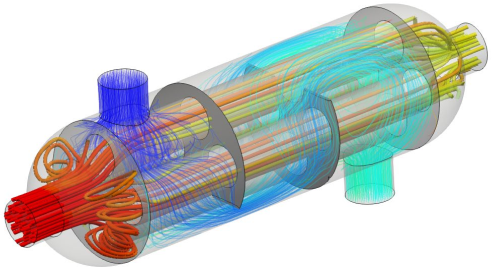
Image credit: https://doi.org/10.3390/pr13061876
https://dx.doi.org/10.1080/09715010.2020.1830000
https://help.sim-flow.com/tutorials/heat-exchanger

Why use CFD?

- **To understand non-uniformities in flow, temperatures and/or concentrations**
  - **Dead zones**
  - **Mixing performance**
  - **Performance heat transfer (cooling pipes, heat exchangers)**

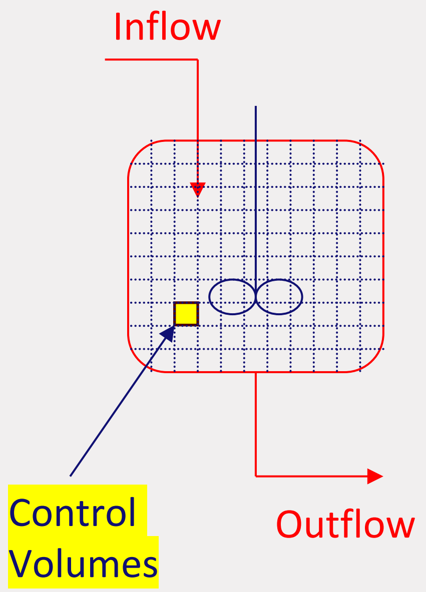






What do we need?

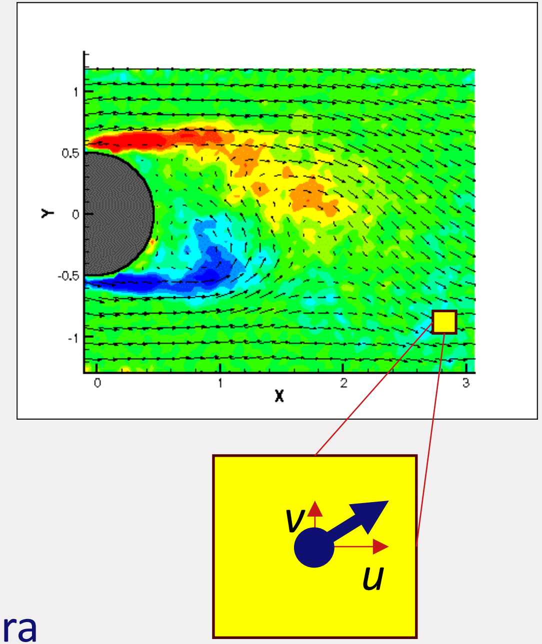
- **A description of primary processes at micro level:** ***microbalances*** ***or*** ***conservation equations***
- **Boundary conditions**
- **Physical and chemical properties of the fluids in the system**
- **Good numerical tools**



Conventions on notation

- **Most theory explained in 1D or 2D (see book* for 3D)**
- **Symbols:**
  - *u, v, w* velocity components
  - *T* temperature
  - Y_j mass fraction component *j* (see lecture 7)
  - ξ mixture fraction (see lecture 7)

- **Computer programs written in Matlab,**
  with references to notation and equations to:
  An Introduction to Computational Fluid Dynamics:
  The Finite Volume Method, H.K. Versteeg, W. Malalasekera

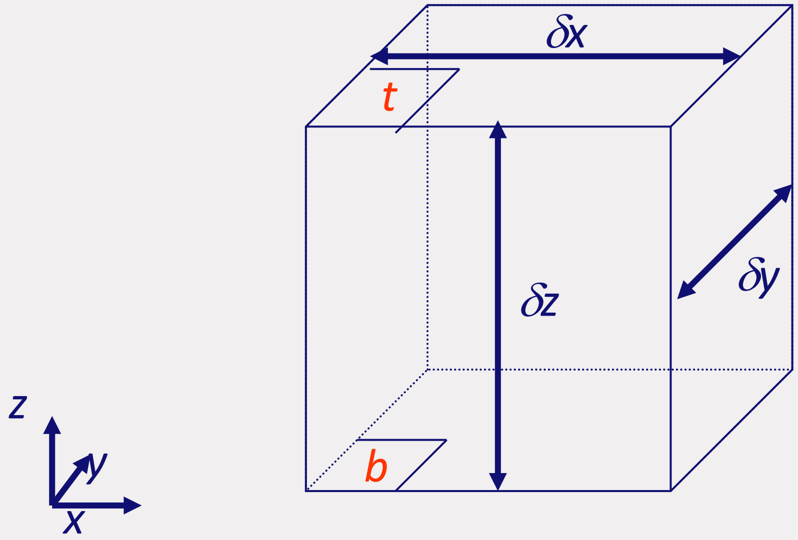


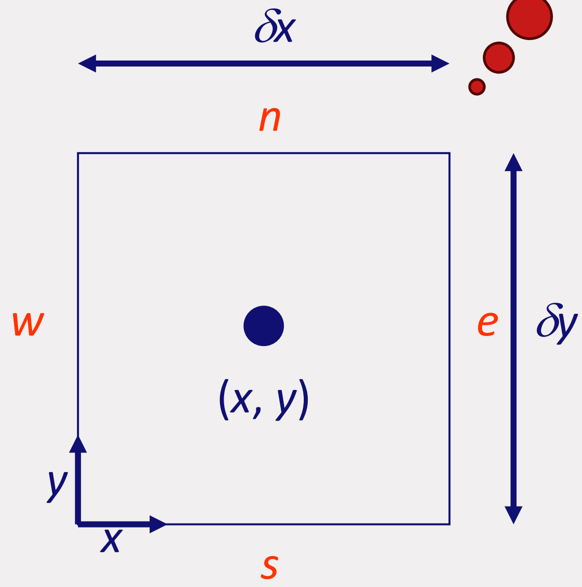
Control volume

Image credit:
https://web1.eng.famu.fsu.edu/~shih/succeed/cylinder/cylinder.htm

Control volume notation

Six faces: *n, s, e, w, t, b*





TU/e

Conservation of mass

- **Accumulation of mass**

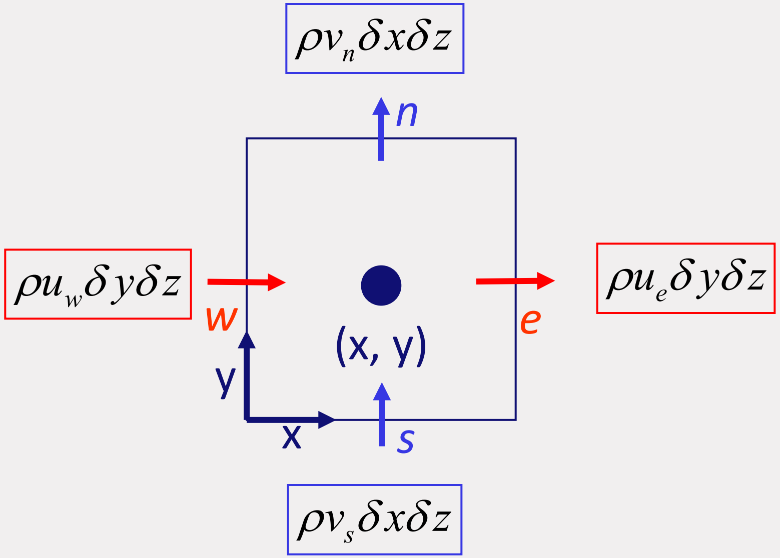
$$\frac{\partial}{\partial t}(\rho \delta x \delta y \delta z) = \frac{\partial \rho}{\partial t}\delta x \delta y \delta z$$

- **Net mass inflow**

$$(\rho u_e - \rho u_w)\delta y \delta z +$$
$$(\rho v_n - \rho v_s)\delta x \delta z$$

- **Divide by control volume ($\delta x\ \delta y\ \delta z$)**

$$\text{Net mass inflow} = \frac{\partial \rho u}{\partial x} + \frac{\partial \rho v}{\partial y}$$



Conservation of mass

- **Microbalance for mass, 2D:**

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} = 0$$

- **Microbalance for mass, 3D:**

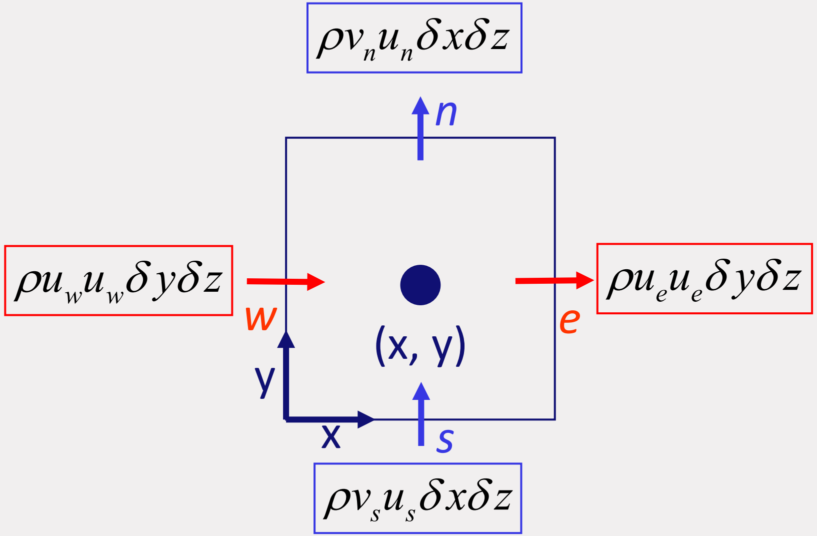
$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} + \frac{\partial (\rho v)}{\partial y} + \frac{\partial (\rho w)}{\partial z} = 0$$

- **General notation with divergence operator div:**

$$\frac{\partial \rho}{\partial t} + \mathrm{div}(\rho \mathbf{u}) = 0$$

Conservation of x-momentum

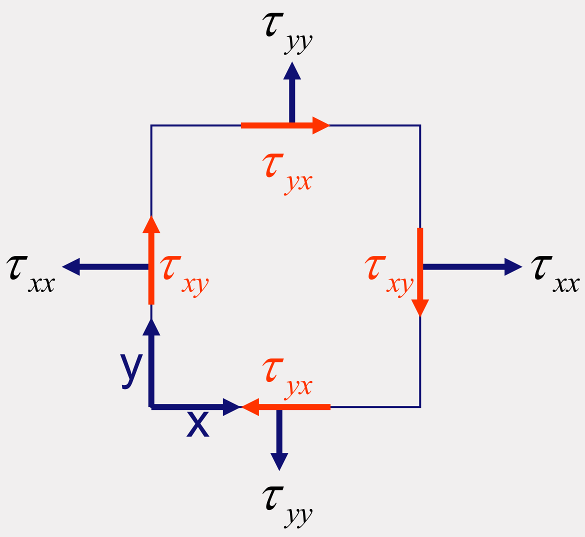
- $\underbrace{\text{Accumulation + net inflow of momentum}}_{\dfrac{\partial \rho u}{\partial t} + \operatorname{div}(\rho u \mathbf{u})}$ = surface forces + body forces



- Surface forces act on the *cell surfaces*
  - Viscous forces
  - Pressure forces
- Body forces act on the *cell volume*
  - Gravity, centrifugal, magnetic, etc.

Surface forces: viscous stress

- **Viscous stress = Normal stress + Shear stress**



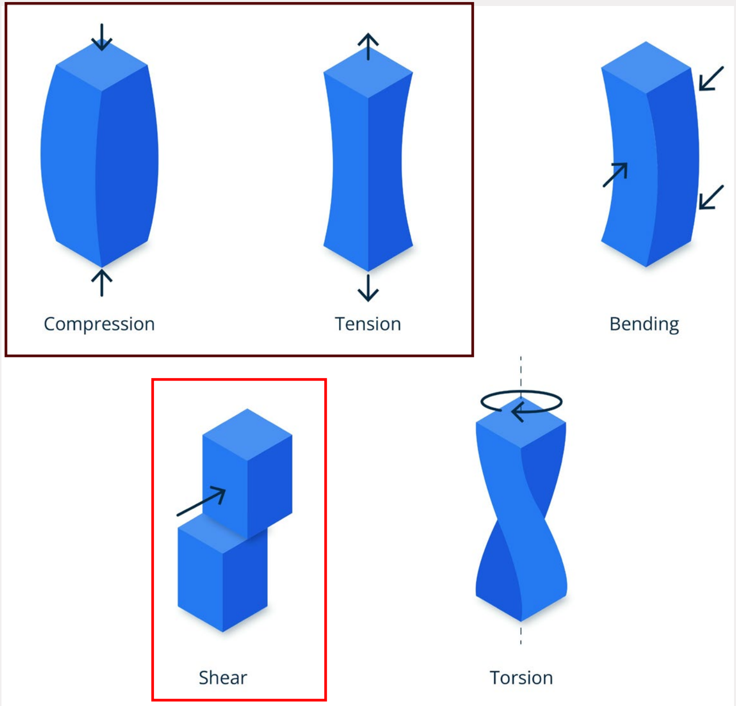


Image credit:
https://www.xometry.com/resources/materials/shear-stress/

Surface forces, x-direction

- **Net surface force on the side faces (e,w):**

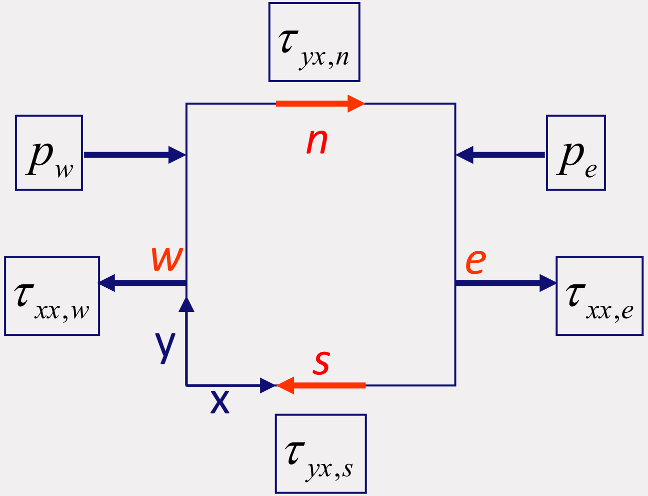
$$(p_w - \tau_{xx,w})\delta y \delta z + (-p_e + \tau_{xx,e})\delta y \delta z = \frac{\partial(-p\tau_{xx})}{\partial x}\delta x \delta y \delta z$$

- **Net surface force on the upper/lower faces (n,s):**

$$-\tau_{yx,w}\delta x \delta z + \tau_{yx,e}\delta x \delta z = \frac{\partial \tau_{yx}}{\partial y}\delta x \delta y \delta z$$

- **Add all faces and divide by control volume ($\delta x\ \delta y\ \delta z$):**

$$\frac{\partial(-p\tau_{xx})}{\partial x} + \frac{\partial \tau_{yx}}{\partial y}$$



Conservation of x- and y-momentum

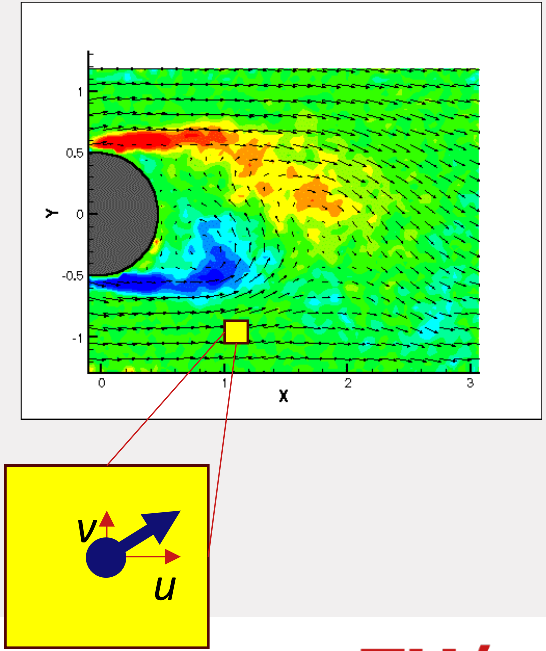
- **Accumulation + net inflow of momentum = surface forces + body forces**

Horizontal: $$\frac{\partial \rho u}{\partial t} + \mathrm{div}(\rho u \mathbf{u}) = \frac{\partial(-p+\tau_{xx})}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + S_{Mx}$$

Vertical: $$\frac{\partial \rho v}{\partial t} + \mathrm{div}(\rho v \mathbf{u}) = \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial(-p+\tau_{yy})}{\partial y} + S_{My}$$

with S_M a body force, for example gravity:

$$S_{Mx} = 0$$
$$S_{My} = -\rho g$$



Control volume

Stress terms

- **Stress terms are function of deformation rate**
- **Stresses in many fluids behave *Newtonian* ($\tau_{xx} \triangleq \frac{\partial u}{\partial x}$) and *isotropic* (no directional dependence)**
- **Stresses in polymers are *non-Newtonian* and *non-isotropic* → not considered here**

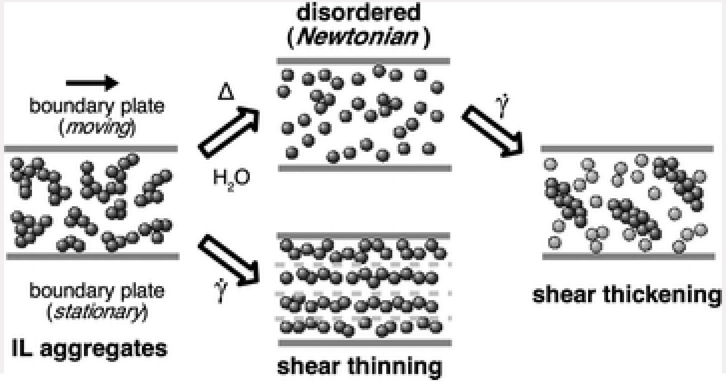


 Image credit: https://doi.org/10.1039/b916049n

Newtonian stress

- **Newtonian stress ≅ deformation rate**
- **μ molecular viscosity:**
- **λ dilatational viscosity:**
  - Gases: λ = -2/3 μ
  - Liquids: div **u** = 0

$$\tau_{xx} = 2\mu \frac{\partial u}{\partial x} + \lambda \,\mathrm{div}\,\mathbf{u}$$

$$\tau_{yy} = 2\mu \frac{\partial v}{\partial y} + \lambda \,\mathrm{div}\,\mathbf{u}$$

$$\tau_{xy} = \tau_{yx} = \mu \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)$$

Conservation of momentum

- **Substituting stress terms gives:**

$$\frac{\partial \rho u}{\partial t} + \operatorname{div}(\rho u \mathbf{u}) = -\frac{\partial p}{\partial x} + \frac{\partial}{\partial x}\left[2\mu\frac{\partial u}{\partial x} + \lambda div\mathbf{u}\right] + \frac{\partial}{\partial y}\left[\mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)\right] + S_{Mx}$$

$$\frac{\partial \rho u}{\partial t} + \operatorname{div}(\rho u \mathbf{u}) = -\frac{\partial p}{\partial x} + \operatorname{div}(\mu \operatorname{grad} u) + S_{Mx}$$

$$\frac{\partial \rho v}{\partial t} + \operatorname{div}(\rho v \mathbf{u}) = -\frac{\partial p}{\partial y} + \frac{\partial}{\partial x}\left[\mu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right)\right] + \frac{\partial}{\partial y}\left[2\mu\frac{\partial v}{\partial y} + \lambda div\mathbf{u}\right] + S_{My}$$

$$\frac{\partial \rho v}{\partial t} + \operatorname{div}(\rho v \mathbf{u}) = -\frac{\partial p}{\partial y} + \operatorname{div}(\mu \operatorname{grad} v) + S_{My}$$

Conservation of energy

- **Derivation of energy equation for incompressible flows leads to:**

$$\rho C_p\left[\frac{\partial T}{\partial t} + \operatorname{div}(T\mathbf{u})\right] = \operatorname{div}(k \operatorname{grad} T) + \underbrace{\tau_{xx}\frac{\partial u}{\partial x} + \tau_{yx}\frac{\partial u}{\partial y} + \tau_{xy}\frac{\partial v}{\partial x} + \tau_{yy}\frac{\partial v}{\partial y}}_{\text{Heating by viscous dissipation} \approx 0} + S_i$$

General transport equation

- **Rate of change + convection = diffusion + (source – sink)**

$$\frac{\partial(\rho\phi)}{\partial t} + \mathrm{div}(\rho\phi\mathbf{u}) = \mathrm{div}(\Gamma\,\mathrm{grad}\,\phi) + S_\phi$$

- **Γ is the diffusion/conduction coefficient**
- **ϕ is a general variable (**1, u, *v*, *w*, *T*, m_j**, …)**
- **Note: diffusion and conduction are analogous processes!**

General transport equation

$$\frac{\partial(\rho\phi)}{\partial t} + \mathrm{div}(\rho\phi\mathbf{u}) = \mathrm{div}(\Gamma\,\mathrm{grad}\,\phi) + S_\phi$$

ϕ	Γ	S_ϕ
1(one)	0	0
u	μ	$-\frac{\partial p}{\partial x} + \frac{\partial}{\partial x}\left[\mu\frac{\partial u}{\partial x} + \lambda\,\mathrm{div}\,\mathbf{u}\right] + \frac{\partial}{\partial y}\left[\mu\frac{\partial v}{\partial x}\right]$
v	μ	$-\frac{\partial p}{\partial y} + \frac{\partial}{\partial x}\left[\mu\frac{\partial u}{\partial y}\right] + \frac{\partial}{\partial y}\left[\mu\frac{\partial v}{\partial y} + \lambda\,\mathrm{div}\,\mathbf{u}\right]$
T	k/C_p	S_i

Integral general transport equation

- To solve the general transport equation we integrate over control volume CV

$$\int_{CV} \frac{\partial(\rho\phi)}{\partial t} dV + \int_{CV} \operatorname{div}(\rho\phi\mathbf{u}) dV = \int_{CV} \operatorname{div}(\Gamma \operatorname{grad} \phi) dV + \int_{CV} S_\phi dV$$

- We use Gauss' divergence theorem

$$\int_{CV} \operatorname{div} \mathbf{a} dV = \int_{A} \mathbf{n} \cdot \mathbf{a} dA$$

$$\frac{\partial}{\partial t}\left(\int_{CV} \rho\phi dV \right) + \int_{A} \mathbf{n} \cdot (\rho\phi\mathbf{u}) dA = \int_{A} \mathbf{n} \cdot (\Gamma \operatorname{grad} \phi) dA + \int_{CV} S_\phi dV$$

Integral general transport equation

$$\frac{\partial}{\partial t}\left(\int_{CV} \rho\phi dV\right) + \int_A \mathbf{n}\cdot(\rho\phi\mathbf{u})dA = \int_A \mathbf{n}\cdot(\Gamma\,\mathrm{grad}\,\phi)dA + \int_{CV} S_\phi dV$$

- **Steady state problems ($\partial/\partial t = 0$):**

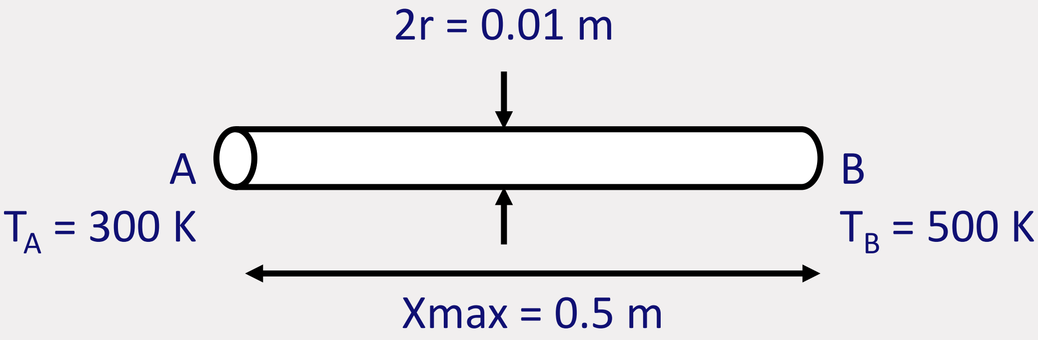
$$\int_A \mathbf{n}\cdot(\rho\phi\mathbf{u})dA = \int_A \mathbf{n}\cdot(\Gamma\,\mathrm{grad}\,\phi)dA + \int_{CV} S_\phi dV$$

- **Time-dependent problems:**

$$\int_{\Delta t}\frac{\partial}{\partial t}\left(\int_{CV} \rho\phi dV\right)dt + \int_{\Delta t}\int_A \mathbf{n}\cdot(\rho\phi\mathbf{u})dAdt =$$

$$\int_{\Delta t}\int_A \mathbf{n}\cdot(\Gamma\,\mathrm{grad}\,\phi)dAdt + \int_{\Delta t}\int_{CV} S_\phi dVdt$$

Example: steady state conduction



Conduction or diffusion problems

- **General transport equation:**

$$\frac{\partial(\rho\phi)}{\partial t} + \mathrm{div}(\rho\phi\mathbf{u}) = \mathrm{div}(\Gamma\,\mathrm{grad}\,\phi) + S_\phi$$

- **Steady diffusion (no convection):**

$$0 = \mathrm{div}(\Gamma\,\mathrm{grad}\,\phi) + S_\phi$$

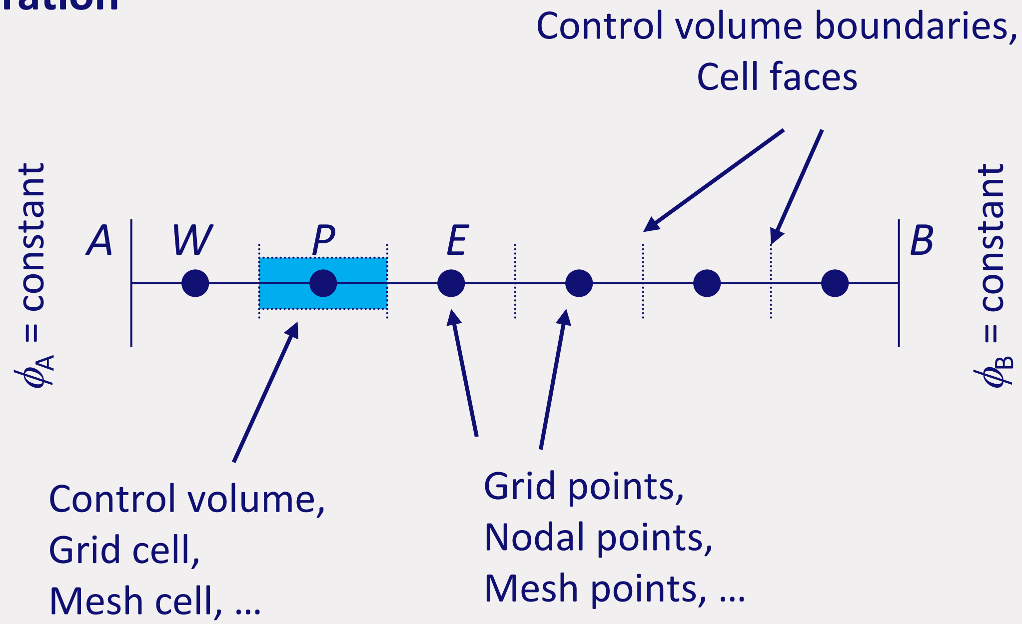
- **1D steady diffusion:**

$$\frac{d}{dx}\left(\Gamma\frac{d\phi}{dx}\right) + S_\phi = 0$$

TU/e

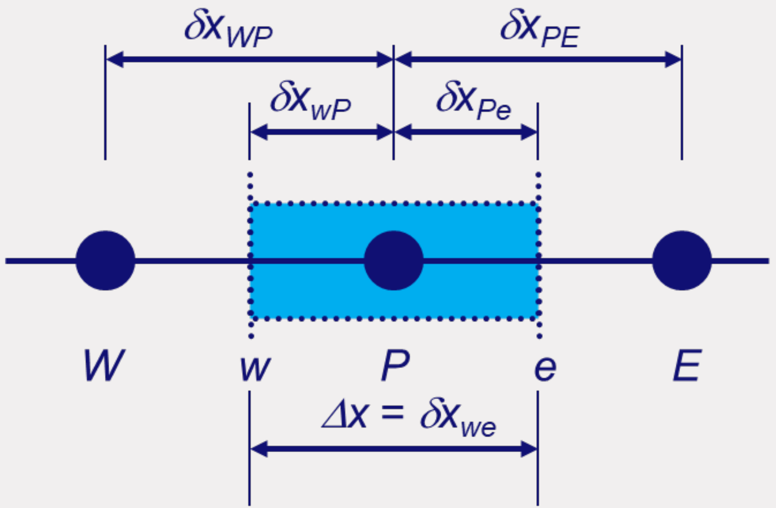
Diffusion problems

Step 1: Grid generation



Diffusion problems

Step 1: Grid generation




```
Dx = XMAX/NPI; % length of volume element

% length variable for the scalar points in the x direction
x     = zeros(1,NPI+2);
x(1) = 0.;
x(2) = 0.5*Dx;
for i = 3:NPI+1
    x(i) = x(i-1) + Dx;
end
x(NPI+2) = x(NPI+1) + 0.5*Dx;
```

Diffusion problems

Step 2: Discretization

$$\int_{\Delta V} \frac{d}{dx}\left(\Gamma \frac{d\phi}{dx}\right) dV + \int_{\Delta V} S_\phi dV = \left(\Gamma A \frac{d\phi}{dx}\right)_e - \left(\Gamma A \frac{d\phi}{dx}\right)_w + \bar{S}\Delta V = 0$$

- **Diffusion coefficients; linear interpolation:**

$$\Gamma_w = \frac{\Gamma_W + \Gamma_P}{2} \qquad \Gamma_e = \frac{\Gamma_P + \Gamma_E}{2}$$

- **Diffusive fluxes:**

$$\left(\Gamma A \frac{d\phi}{dx}\right)_e = \Gamma_e A_e \left(\frac{\phi_E - \phi_P}{\delta x_{PE}}\right) \qquad \left(\Gamma A \frac{d\phi}{dx}\right)_w = \Gamma_w A_w \left(\frac{\phi_P - \phi_W}{\delta x_{WP}}\right)$$

Diffusion problems

- **Source term linearization:**

$$\bar{S}\Delta V = S_u + S_P\phi_P$$

- **Substitute diffusive fluxes and source term:**

$$\Gamma_e A_e\left(\frac{\phi_E-\phi_P}{\delta x_{PE}}\right)-\Gamma_w A_w\left(\frac{\phi_P-\phi_W}{\delta x_{WP}}\right)+(S_u+S_P\phi_P)=0$$

- **Rearrange:**

$$\boxed{\left(\frac{\Gamma_e}{\delta x_{PE}}A_e+\frac{\Gamma_w}{\delta x_{WP}}A_w-S_P\right)\phi_P=\left(\frac{\Gamma_w}{\delta x_{WP}}A_w\right)\phi_W+\left(\frac{\Gamma_e}{\delta x_{PE}}A_e\right)\phi_E+S_u}$$

Diffusion problems

$$\left(\frac{\Gamma_e}{\delta x_{PE}}A_e + \frac{\Gamma_w}{\delta x_{WP}}A_w - S_P\right)\phi_P = \left(\frac{\Gamma_w}{\delta x_{WP}}A_w\right)\phi_W + \left(\frac{\Gamma_e}{\delta x_{PE}}A_e\right)\phi_E + S_u$$

- **Introduce coefficients** a_P , a_W , a_E

$$\boxed{a_P\phi_P = a_W\phi_W + a_E\phi_E + S_u}$$

- **where**

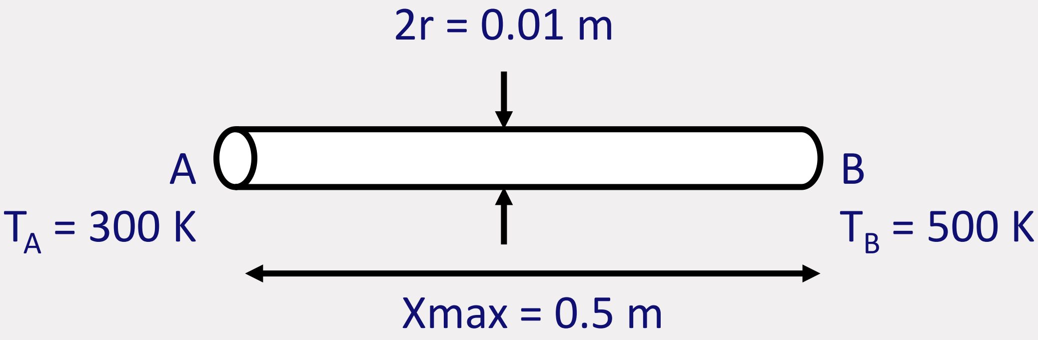
a_W	a_E	a_P
$\frac{\Gamma_w}{\delta x_{WP}}A_w$	$\frac{\Gamma_e}{\delta x_{PE}}A_e$	$a_W + a_E - S_P$

Diffusion problems

Step 3: Solution of equations

- **An equation solver is treated in HC 3**
- **Solver considered a black box for the moment**

Example: steady state conduction



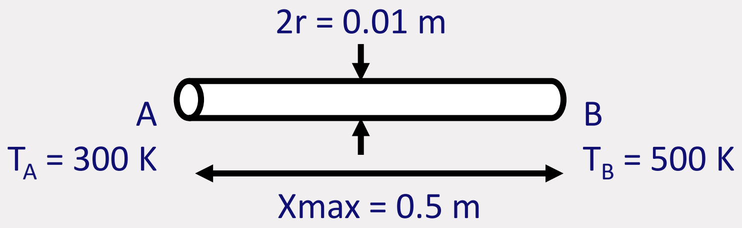
Numerical solution

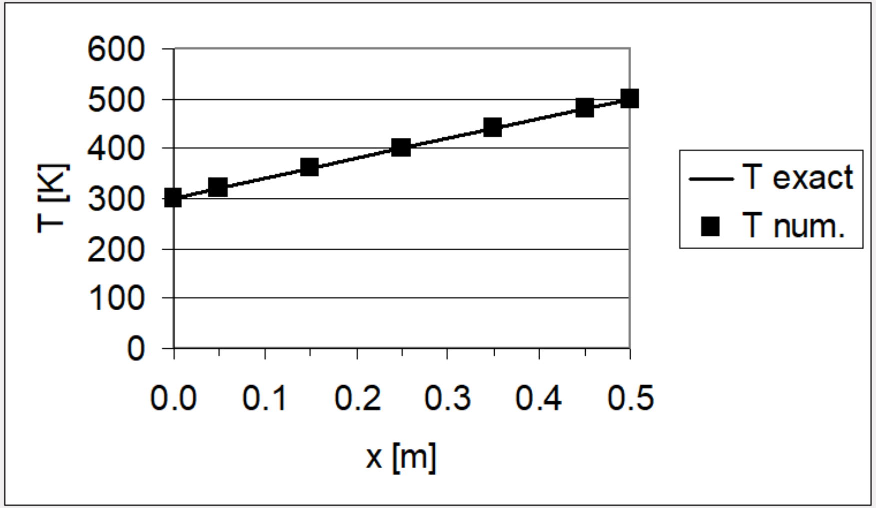
```
%% CONSTANTS          /* define all the constants */
%% INIT               /* initialize variables */
%% BOUNDARY           /* apply boundary conditions */
%% T-EQUATION         /* calculate coefficients of T equation aE, aW, aP, b */
%% SOLVE              /* solver iteration loop */
    for iter = 0:OUTER_ITER
        …             /* solve1D(T, b, aE, aW, aP)*/
        fprintf(…);   /* write convergence to screen */
    end
%% OUTPUT             /* write output to file */
```

Output on the command window

```
 0 T[1] = 305.556 T[2] = 316.667 T[3] = 333.333 T[4] = 366.667 T[5] = 433.333
 1 T[1] = 311.121 T[2] = 333.362 T[3] = 361.169 T[4] = 402.894 T[5] = 462.731
 2 T[1] = 314.931 T[2] = 344.792 T[3] = 378.463 T[4] = 420.780 T[5] = 472.042
 3 T[1] = 317.201 T[2] = 351.602 T[3] = 388.274 T[4] = 429.851 T[5] = 475.963
 4 T[1] = 318.477 T[2] = 355.431 T[3] = 393.661 T[4] = 434.585 T[5] = 477.876
 5 T[1] = 319.177 T[2] = 357.530 T[3] = 396.583 T[4] = 437.097 T[5] = 478.867
 6 T[1] = 319.556 T[2] = 358.668 T[3] = 398.160 T[4] = 438.441 T[5] = 479.393
 7 T[1] = 319.761 T[2] = 359.283 T[3] = 399.010 T[4] = 439.162 T[5] = 479.674
 8 T[1] = 319.871 T[2] = 359.614 T[3] = 399.467 T[4] = 439.549 T[5] = 479.825
 9 T[1] = 319.931 T[2] = 359.792 T[3] = 399.713 T[4] = 439.757 T[5] = 479.906
10 T[1] = 319.963 T[2] = 359.888 T[3] = 399.846 T[4] = 439.869 T[5] = 479.949
11 T[1] = 319.980 T[2] = 359.940 T[3] = 399.917 T[4] = 439.930 T[5] = 479.973
12 T[1] = 319.989 T[2] = 359.968 T[3] = 399.955 T[4] = 439.962 T[5] = 479.985
13 T[1] = 319.994 T[2] = 359.983 T[3] = 399.976 T[4] = 439.980 T[5] = 479.992
14 T[1] = 319.997 T[2] = 359.991 T[3] = 399.987 T[4] = 439.989 T[5] = 479.996
15 T[1] = 319.998 T[2] = 359.995 T[3] = 399.993 T[4] = 439.994 T[5] = 479.998
16 T[1] = 319.999 T[2] = 359.997 T[3] = 399.996 T[4] = 439.997 T[5] = 479.999
17 T[1] = 320.000 T[2] = 359.999 T[3] = 399.998 T[4] = 439.998 T[5] = 479.999
18 T[1] = 320.000 T[2] = 359.999 T[3] = 399.999 T[4] = 439.999 T[5] = 480.000
19 T[1] = 320.000 T[2] = 360.000 T[3] = 399.999 T[4] = 440.000 T[5] = 480.000
20 T[1] = 320.000 T[2] = 360.000 T[3] = 400.000 T[4] = 440.000 T[5] = 480.000
```

Comparison with analytical results





2D and 3D problems

- **Step 1: Grid generation**
  - **Six faces: n, s, e, w, t, b**
- **Step 2: Discretization**
  - $\Delta V = \delta x \, \delta y \, \delta z$
- **Step 3: Solution of equations**
  - **Treated in HC 3 or Chapter 6**
- **See book for detailed description**

Wrap up

- Rate of change + convection = diffusion + (source – sink)

$$\frac{\partial(\rho\phi)}{\partial t} + \mathrm{div}(\rho\phi\mathbf{u}) = \mathrm{div}(\Gamma\,\mathrm{grad}\,\phi) + S_\phi$$

- General form of discretized equations:

$$\boxed{a_P\phi_P = \sum a_{nb}\phi_{nb} + S_u} \qquad \boxed{a_P = \sum a_{nb} - S_P}$$

- Source terms are included through S_u and S_P

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

Introduction to CFD (4RC30)

Basics of programming in Matlab

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Department of Mechanical Engineering

Variables

- **Script/Function file: .m extension**

```
function [] = init()
% This function is to initialise all parameters.
```

- **Variables: every variable is an array or matrix.**

```
%% constants
NPI       = 5;                % number of grid cells in x-direction [-]
XMAX      = 1.0;              % width of the domain [m]
Dx        = XMAX / NPI;       % length of volume element
x         = zeros(1,NPI+2);   % x coordinate on pressure points [m]
r         = [7 8 9 10 11];    % row vectors
c         = [7;  8;  9;  10; 11]; % column vectors
```

Operators

- ## Arithmetic operators

Symbol	Role
+	Addition
-	Subtraction
.*	Element-wise multiplication
*	Matrix multiplication
./	Element-wise right division
/	Matrix right division
.^	Element-wise power
^	Matrix power

Relational operators

Symbol	Role
==	Equal to
~=	Not equal to
>	Greater than
>=	Greater than or equal to
<	Less than
<=	Less than or equal to

Logical operators

<table>
<tr><th>Symbol</th><th>Role</th></tr>
<tr><td>&</td><td>Logical AND</td></tr>
<tr><td>|</td><td>Logical OR</td></tr>
<tr><td>&&</td><td>Logical AND (with short-circuiting)</td></tr>
<tr><td>||</td><td>Logical OR (with short-circuiting)</td></tr>
<tr><td>~</td><td>Logical NOT</td></tr>
</table>

- ## Functions

Function	Explanation
sum(A,dim)	Sums along the dimension of *A* specified by scalar *dim*.
ceil(A)	rounds the elements of A to the nearest integers greater than or equal to A.
floor(A)	rounds the elements of A to the nearest integers less than or equal to A.
Round(A)	rounds the elements of A to the nearest integers.

Decisions (if statement)

- **if … end statement**

```
if <expression>
        <statement(s)>
end
```

```
a = 10;
if a < 20 % check the condition
   % if condition is true then print the following
    fprintf('a is less than 20\n' );
end
```

a is less than 20

- **if… elseif … else … end statement**

- **if … else … end statement**

```
if <expression>
              <statement(s)>
else
              <statement(s)>
end
```

```
a = 100;
if a < 20   % check the boolean condition
    % if condition is true then print the following
    fprintf('a is less than 20\n' );
else
    % if condition is false then print the following
    fprintf('a is not less than 20\n' );
end
```

a is not less than 20

Loop

- **for … end statement**

```
for index = values
        <program statements>
        ...
end
```

```
for i = 1:NPI+2
      fprintf(fp,'%11.4e\t%11.4e\n',x(i),T(i));
end
```

- **while… end statement**

```
while <expression>
        <statements>
        ...
end
```

```
a = 10;
% while loop execution
while( a < 20 )
      fprintf('value of a: %d\n', a);
      a = a + 1;
end
```

Import and Output

- **Data import**

A = importdata(filename)
% Loads data into array A from the file denoted by filename.

```
filename = ‘output.dat';
A = importdata(filename);
```

- **Output**

Many options

```
% write output to txt file
if iter == OUTER_ITER
    fp = fopen('output.txt','w');
    for i = 1:NPI+2
        fprintf(fp,'%11.4e\t%11.4e\n',x(i),T(i));
    end
    fclose(fp);
end
```

Plotting & Images

Plot

```
x = [0:5:100];
y = x;
plot(x, y)
```

Contour

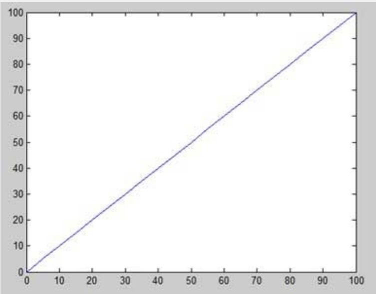
```
[x,y] = meshgrid(-5:0.1:5,-3:0.1:3);
g = x.^2 + y.^2;
contour(x,y,g)
```

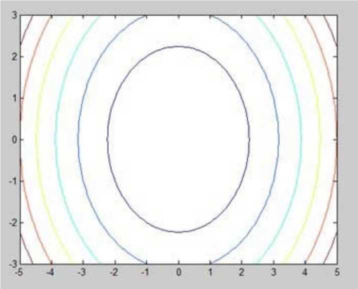
3D plot

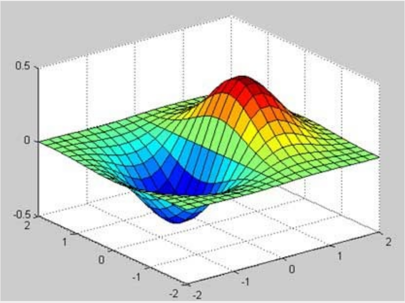
```
[x,y] = meshgrid(-2:.2:2);
g = x .* exp(-x.^2 - y.^2);
surf(x, y, g)
```

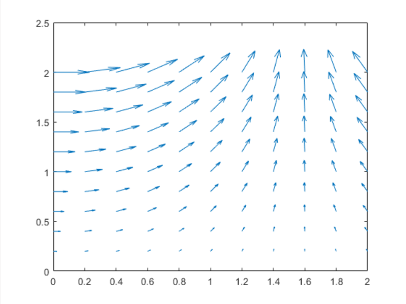
Vector field

```
[x,y] = meshgrid(0:0.2:2,0:0.2:2);
u = cos(x).*y;
v = sin(x).*y;
quiver(x,y,u,v)
```









Wellbeing Signal Group @ ME

INTRODUCTION SLIDES FOR COURSES

Diversity statement of ME

“The Mechanical Engineering department at TU/e is committed to creating an inclusive and welcoming environment that values diversity and promotes equity.

We strive to create a community where everyone feels safe, respected, and valued.

We believe that diversity is essential to our mission of educating students to become global citizens who can thrive in an increasingly diverse world.

We are committed to fostering an environment where all members of our community - students, faculty, and staff - can learn, grow, and succeed.”

— Dean prof.dr.ir. P. D. Anderson

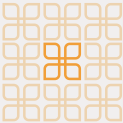
Goal of the Wellbeing Signal Group @ ME

… reducing barriers for conversations!

Our goal is to ensure that all individuals are supported on their journey to thrive, both professionally and personally.

We help picking up signals affecting your wellbeing as small as they might be. We take those signals, which usually reveal a pattern, and we take actions accordingly.

Signals



Patterns

Actions/Response-ability

Promotors of wellbeing: Meet your representatives

Monica Zakhari, Chair

Tom Oomen, Full professors representative

Idoia Cortes Garcia, YRN representative

Apoorva Sonawane, ESA representative

Hind Boulgalag, HR representative

Toos van Gool, PhD representative

Claudia Hanegraaf, Masters

Elise van Lieshout, Masters

Lucas Haselhoff, Bachelor

Raghad Benothman, Bachelor

Student representatives

https://www.tue.nl/en/our-university/departments/mechanical-engineering/wellbeing

Feel free to contact us if you want to see something changed.
You can do so personally, through the website, or by email:
WellbeingSG-ME@tue.nl

TU/eMechE TU/e

Variables at cell faces

- **Control volume (grid cell, fluid element) is very small**
- **Taylor series for pressure at east face:**

- **Only use first two terms of Taylor series**

$$p(x+\tfrac{1}{2}\delta x) = p(x) + \frac{\partial p(x)}{\partial x}\tfrac{1}{2}\delta x + \tfrac{1}{2}\frac{\partial^2 p(x)}{\partial x^2}(\tfrac{1}{2}\delta x)^2 + \ldots$$

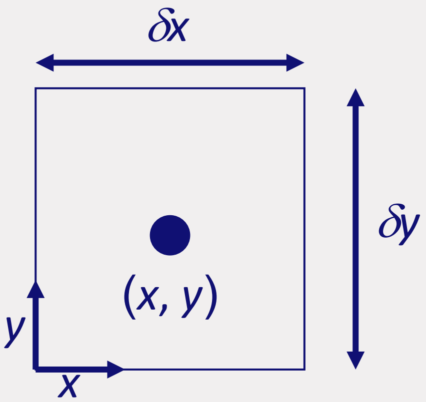
Conservation of mass

- **Accumulation of mass + Net mass inflow = 0**
- **Accumulation:**

$$\frac{\partial}{\partial t}(\rho \delta x \delta y) = \frac{\partial \rho}{\partial t}\delta x \delta y$$

- **Divide by area control volume ($\delta x \ \delta y$):**

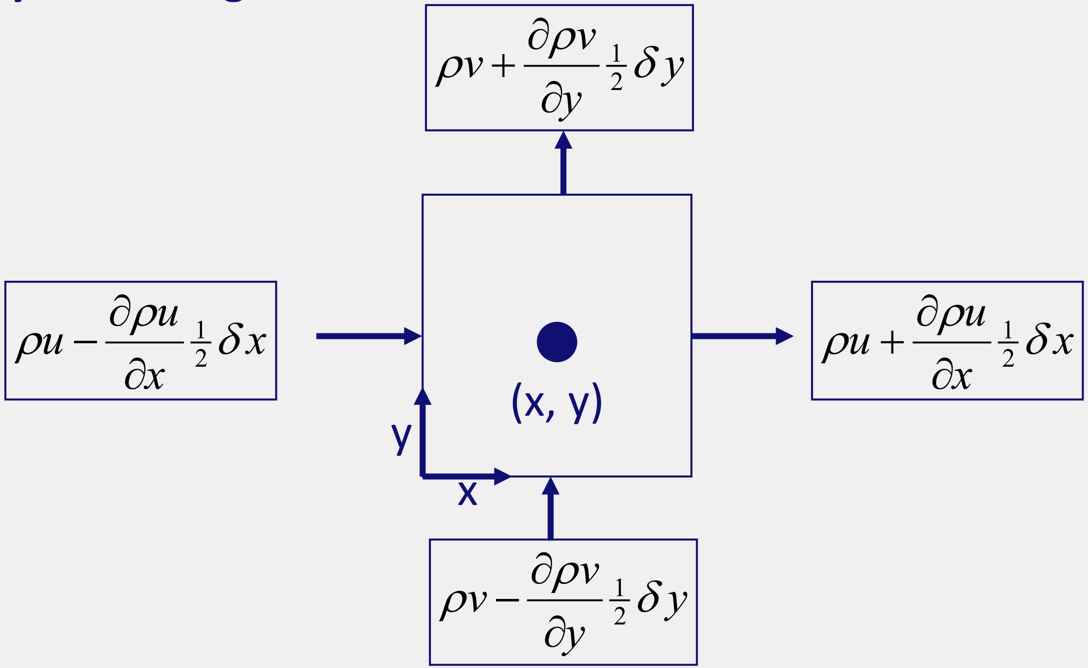
$$\frac{\partial \rho}{\partial t}$$



TU/e

Conservation of mass

- **Taylor series gives for net mass inflow at faces:**



Conservation of mass

- **Add up all mass flows into control volume:**

$$\left(\rho u - \frac{\partial \rho u}{\partial x}\tfrac{1}{2}\delta x\right)\delta y - \left(\rho u + \frac{\partial \rho u}{\partial x}\tfrac{1}{2}\delta x\right)\delta y$$

$$+\left(\rho v - \frac{\partial \rho v}{\partial y}\tfrac{1}{2}\delta y\right)\delta x - \left(\rho v + \frac{\partial \rho v}{\partial y}\tfrac{1}{2}\delta y\right)\delta x$$

- **Divide by area control volume ($\delta x\ \delta y$):**

$$\text{Net mass inflow} = \frac{\partial \rho u}{\partial x} + \frac{\partial \rho v}{\partial y}$$

Conservation of mass

- **Microbalance for mass, 2D:**

$$\frac{\partial\rho}{\partial t}+\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}=0$$

- **Microbalance for mass, 3D:**

$$\frac{\partial\rho}{\partial t}+\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0$$

$$\frac{\partial\rho}{\partial t}+\mathrm{div}(\rho\mathbf{u})=0$$